WEIR - Assignment 3 - Retrieval-Augmented Generation

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***Abstract*—This report describes a Retrieval-Augmented Generation (RAG) system built on top of the PA2 document retrieval pipeline from the previous assignment. The system answers user questions in two modes: with context (retrieval, reranking and generation) and without context (LLM-only). We tested the results of these modes across multiple models and example queries using a judge model and citation-reference validation.**

I. Introduction

This assignment extends our PA2 retrieval system with a language model to create a RAG pipeline that can answer questions in two modes: with context, where the model receives retrieved document chunks as explicit evidence, and without context, where the model answers directly from its parameters.

II. Implementation Summary

A. Models and tooling

Our system consists of three model components: a sentence embedding model for retrieval, a reranker to refine candidates, and a large language model (LLM) for answer generation. We experimented with multiple models for both embedding and reranking. For answer generation, we use a locally served model via Ollama and interface with the LLM through `DSPy`.

B. Document storage, chunking and chunk filtering

We reuse the PA2 database schema and chunked documents.

To mitigate noise from very short segments (e.g., headers, keywords, or stray fragments that clutter the vector space), retrieval incorporates a minimum segment length filter set at 100 characters by default. This filter is applied during similarity search: both the approximate nearest neighbor candidate selection and the final ranking stage exclude segments shorter than the threshold, ensuring that only substantive text blocks are candidate for evidence.

C. Retrieval

For answering with context, the pipeline embeds the user query, retrieves the top-k most similar segments from pgvector using a cosine distance metric and reranks the candidates with a cross-encoder. The final top-n segments are formatted into a structured context block that is provided to the LLM.

D. Prompt construction

We use two prompting regimes.

With context prompts are designed for strict evidence-based answering. Using a system prompt, they constrain the model to use only the provided context (no external knowledge), to keep answers concise (approximately 40-110 words and at most four sentences), and to support each important factual claim with an inline citation using a chunk identifier (e.g., `[ChunkID: <id>]`). In addition to the natural-language answer, the model must return a structured reference list in JSON format, that matches the cited identifiers exactly. The reference list must be a JSON array of objects with fields `chunk_id` and `url`, e.g., `[{"chunk_id": "<id>", "url": "<url>"}]`. References that are not cited in the answer are disallowed.

Within this with context setting, we evaluate two prompting styles: zero-shot and one-shot. In the zero-shot variant, the system prompt consists only of the instruction set (grounding rules, citation format, fallback behavior, and the JSON reference constraints). In the one-shot variant, we prepend a single fully worked example (example question, example context with chunk blocks, and an example output containing both a cited answer and a valid `references_json` list). This example is not part of the retrieved evidence for the actual question; instead, it serves as an in-prompt demonstration of the expected formatting and of the strict one-to-one correspondence between in-text ChunkID citations and the reference list. The text template defining the instruction set and the example can be read in Appendix A.

If the context is insufficient to answer the question, the model must output the exact fallback message:

```
Na podlagi podanega konteksta
tega ni mogoče ugotoviti.
```

In this case, the structured reference list must be empty.

Without context prompts provide an LLM-only baseline. They prohibit citations and source reporting and encourage answers from general knowledge. To prevent fabricated sourcing, citations, identifiers, and URLs are explicitly disallowed. If the question clearly requires missing document-specific information, the model must return the exact fallback message:

```
Brez dodatnega konteksta tega ni mogoče
zanesljivo ugotoviti.
```

E. Explainable context formatting and validation

Retrieved evidence is formatted as human-readable blocks that are passed to the LLM as the context. Each block includes a local header, a segment identifier used for citations, the page URL, and the chunk text. Including URLs directly in the context reduces the need for the model to invent sources and supports the structured reference output.

Below we include the template used to construct the evidence context for a specific chunk (values in braces are filled in at runtime):

```
### ODSEK {i}
ChunkID: {chunk_id}
URL: {page_url}
Besedilo:
{text}

---------------
```

III. Evaluation procedure

We implemented a systematic three-stage evaluation framework centered on two curated datasets.

A. Datasets

The *biased* evaluation dataset comprises 44 queries constructed via a supervised process using `GPT-4.1-mini` (via OpenAI API) as a question and expected-answer generator. For each sampled page, chunk windows are selected and `GPT-4.1-mini` generates a factual question in Slovenian, a concise expected answer, and a set of atomic key facts. Each fact is explicitly linked to supporting chunk identifiers. Candidate chunks are then annotated as required (minimal support set), acceptable (alternative support), partial (incomplete support), or hard negatives (unrelated). The dataset is validated algorithmically: all chunk IDs are verified to exist on their pages, and malformed or unsupported examples are rejected.

Example: One dataset query asks "*Koliko gasilcev in vojakov je rusko ministrstvo za izredne razmere že napotilo v boj proti požarom, koliko dodatnih vojakov je vpoklical današnji odlok Medvedjeva, koliko smrtnih žrtev in koliko uničenega ozemlja so povzročili požari v zahodni Rusiji ter kateri vremenski pogoji so glavni vzrok za požare?*". The expected answer references four key facts, described in a single chunk: deployment of hundreds of thousands of firefighters and approximately 2000 military personnel, Medvedev's decree calling additional troops, 40 deaths and 128,500 hectares destroyed, hot and dry weather as root cause.

The second *unbiased* dataset was generated directly by ChatGPT 5.5 by prompting it to generate 20 different questions on a mixed topic of Donald Trump, Iran, Ukraine, and Israel, then answered in at most 4 sentences, and provided with 5 supported atomic facts. Citations here are also disallowed since the model cannot know about our database of chunks.

B. Retrieval Evaluation

Retrieval quality was remeasured using Normalized Discounted Cumulative Gain at multiple cut-offs. The metric is computed against LLM-generated ground truth scores and cross-encoder rerankings. We evaluate multiple configuration combinations of the `bge-m3` [1] embedding cosine distance retrieval paired with different reranking strategies (e.g. no reranking, `ms-marco-MiniLM-L6-v2` [2], `mmarco-mMiniLMv2-L12-H384-v1` [3], `bge-reranker-v2-m3` [4]).

C. Citation Quality Evaluation

For evidence-based answering, we perform bidirectional citation-reference validation: every chunk identifier cited in the answer (via inline `[ChunkID: <id>]` markers) must appear in the structured reference JSON list, and vice versa. Additional metrics include precision and recall against the required and acceptable chunk sets (where precision = relevant citations / total citations, and recall = relevant citations / acceptable chunk set size), and detection of hard-negative contamination (citing chunks marked as unrelated in the dataset). This evaluation quantifies whether the system maintains faithful grounding between in-text citations and references, enforcing that if a fact is stated, it must be cited and if a citation is made, the cited chunk must support it.

D. Answer Quality Evaluation

Answer semantic quality is assessed using GPT-4 as a judge. For each query, the judge compares the model's answer against the standard expected answer and key facts using a detailed rubric. Evaluation dimensions include key-fact coverage (marked as covered, implied, partial, missing, or contradicted), overall semantic score (0.0–1.0 scale), direct question-answer alignment (true/false), and presence of major errors. The rubric is flexible: paraphrases are accepted, citation formatting does not affect semantic scoring, and factually correct supplementary information is tolerated.

Example: Consider the query "*Kako je Donald Trump po vrnitvi na položaj spremenil pristop ZDA do vojne med Rusijo in Ukrajino?*". The expected answer identifies five key facts: Trump's advocacy for faster negotiations, questioning long-term U.S. military aid to Ukraine, emphasis on burden-sharing among European allies, critics' concern about territorial concessions, supporters' view on accelerated peace talks. An LLM model that answered "*Donald Trump ni v trenutno v predsedovanju ZDA, zato ni mogoče ugotoviti, kako bi spremenil pristop do vojne med Rusijo in Ukrajino...*" received a semantic score of 0.0 and key-fact coverage score of 0.0, with all five facts marked as *missing*. The judge's reasoning noted "*Odgovor ne odgovarja na vprašanje, saj zanika možnost ocene Trumpovega pristopa po vrnitvi na položaj in ne pokrije nobenega ključnega dejstva...*" This example illustrates the judge's treatment of factually incorrect or question-misaligned responses.

IV. Evaluation Findings

A. Retrieval Performance and Model Selection

Our retrieval evaluation assessed 50 queries across multiple embedding and reranking configurations. For the `bge-m3` embedding model without reranking, we observe strong early-ranking performance: NDCG@2 = 0.983 (std 0.056), NDCG@3 = 0.917 (std 0.100), with gradual degradation to NDCG@10 = 0.760 (std 0.134). This indicates that relevant chunks are typically placed within the top two to three positions, with degradation as ranking depth increases.

Reranking with `ms-marco-MiniLM-L6-v2` yields comparable early performance (NDCG@2 = 0.973, NDCG@3 = 0.918) but marginally improves mid-range rankings (NDCG@6 through NDCG@10 all exceed no-reranker scores by about 0.003-0.015).

Comparative performance across model and reranker combinations is visualized in Figures 1 and 2. The mean NDCG scores across all cut-offs show consistent patterns: `bge-m3` with no reranking achieves competitive performance, while reranked variants (`ms-marco`, `mmarco`, `bge`) cluster at slightly higher means. Standard deviations remain elevated across configurations, indicating case-by-case variance in retrieval effectiveness, suggesting that no single configuration dominates uniformly.

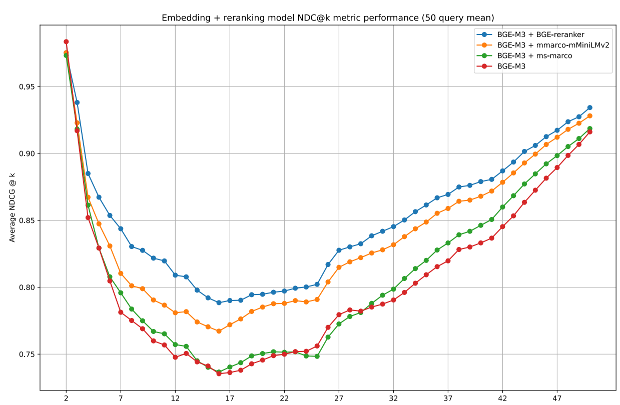

Fig. 1: Mean NDCG@k across embedding and reranking configurations. Bars represent average scores over 50 test queries.

B. Retrieval-Agnostic Baseline: Model Size, Answer Quality

We evaluated three Qwen model sizes in direct (no-context) mode to isolate language capability from retrieval grounding.

As mentioned above, each question includes five key facts extracted from expected answers. For each fact, an external LLM judge (`GPT-4`) assigns a coverage status: `covered` (1.0), `implied` (0.85), `partial` (0.5), `missing` (0.0), or `contradicted` (0.0). The key-fact coverage score is the arithmetic mean of individual fact scores across all facts. This metric favors exact coverage but maintains partial credit for implicit or partial information.

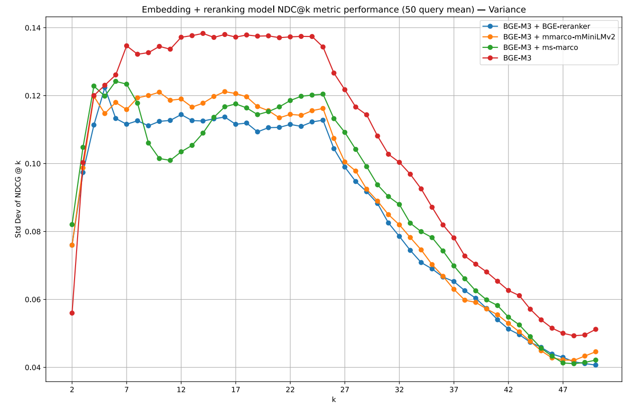

Fig. 2: Standard deviation of NDCG@k across configurations.

The external judge rates overall answer quality on a 0.0–1.0 scale based on five rubric dimensions: directness and clarity in addressing the question, comprehensiveness relative to expected answer, freedom from major factual errors, presence of unsupported claims, and coherence and readability. The judge assigns scores at 0.0-1.0 multiples of 0.2 (e.g., 0.0, 0.2, 0.4, 0.6, 0.8, 1.0).

The final answer quality score combines key fact coverage and semantic quality as: $\text{score}_{\text{composite}} = 0.65 \cdot \text{coverage} + 0.35 \cdot \text{semantic}$. The 0.65/0.35 weighting emphasizes factual grounding but recognizes fluency and semantic appropriateness. Penalties apply via multiplication by 0.5 for three conditions: answer contains contradictions, answer contains major errors, answer does not address the question. Multiple violations compound multiplicatively (e.g., contradicted answer that also doesn't address question: $\text{score}_{\text{final}} = \text{score}_{\text{composite}} \times 0.5 \times 0.5 = 0.25 \times \text{score}_{\text{composite}}$).

Per-example metrics are aggregated across 20 test examples. Mean coverage and semantic scores are simple arithmetic means across examples.

Results confirm clear model-size sensitivity:

- **Qwen 4B** [5]: Mean key-fact coverage 0.486, semantic score 0.58, composite quality 0.516. Hallucination rate 30%, contradiction rate 0%, major-error rate 10%, question-address rate 90%.
- **Qwen 8B** [6]: Mean key-fact coverage 0.471, semantic score 0.56, composite quality 0.497. Hallucination rate 45%, contradiction rate 0%, major-error rate 15%, question-address rate 90%.
- **Qwen 14B** [7]: Mean key-fact coverage 0.549, semantic score 0.67, composite quality 0.581. Hallucination rate 60%, contradiction rate 0%, major-error rate 10%, question-address rate 95%.

The 14B model achieved 13% higher fact coverage (0.549 vs 0.486) and 16% higher semantic score (0.67 vs 0.58) compared to the 4B model, despite 2× higher hallucination rate (60% vs 30%). This trade-off reflects a broader pattern in language models: larger models generate more fluent, paraphrased, and contextually adaptive answers, but simulta-

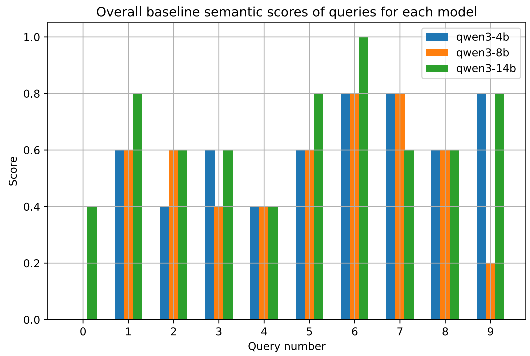

Fig. 3: Semantic scores plotted for first 10 queries of the unbiased dataset for each of the 3 `qwen3` models.

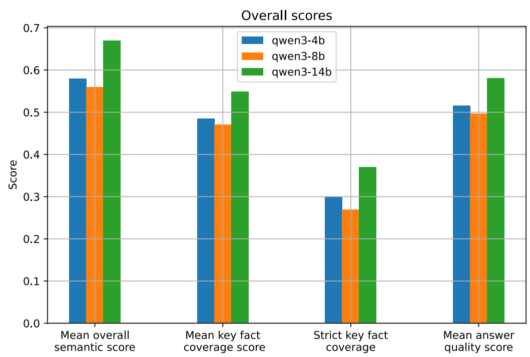

Fig. 4: Overall semantic scores for each of the 3 `qwen3` models on the unbiased dataset.

neously produce more fabricated details when grounding is unavailable. Contradiction rates remained near zero across all sizes, indicating that direct LLM-only answers avoid factual inconsistency with expected answers when semantic content exists. A short summary of certain metrics for all three models is shown in Figure 4.

To better understand in which scenarios certain models perform better or worse, a per query plot in Figure 3 was created, where in some cases `qwen3-14b` outperforms others by 0.2 to 0.4 in score. For example, in the first query `qwen3-4b` stated that Donald Trump is currently (in 2026) not the president so he didn't influence the Russia-Ukraine conflict that much, `qwen3-8b` refused to answer due to a lack of context needed, but `qwen3-14b` managed to put together an answer that covered a few of the required key facts (semantic score was greater than 0). In the 9th query, `qwen3-8b` performed much worse than other models because it focused on enumerating countries for the major part of the answer and didn't cover enough key facts for a higher semantic score.

C. Key-Fact Coverage Breakdown

Across the 14B baseline, out of 100 assessed key facts: 37 were directly covered, 14 implied through paraphrase, 12

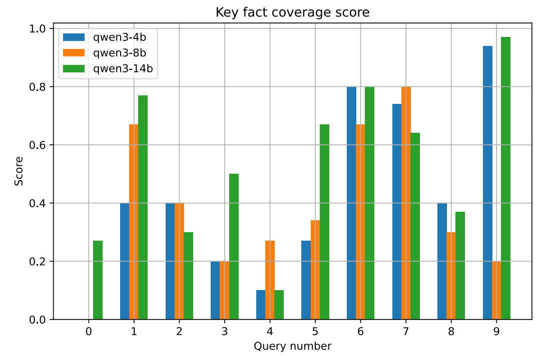

Fig. 5: Key fact coverage scores plotted for first 10 queries of the unbiased dataset for each of the 3 `qwen3` models.

partially addressed with missing components, 36 completely missing, and 1 contradicted. The tight clustering of strict (covered-only) coverage (0.37 vs composite 0.549) indicates that models struggle to comprehensively address all components of multi-part questions, but often convey partial or inferred understanding. The absence of contradictions under no-context evaluation suggests the fallback mechanism is reliable: when uncertain, models correctly default to uncertain rather than assert falsehoods.

Key-fact coverage was also plotted in the Figure 5 for the first 10 queries to check any unexpected results. The shape of the plot is similar to the one in Figure 3 since key fact coverage and semantic score are very linked.

D. Question-Alignment and Error Patterns

Question-address rates of 90-95% suggest that models understand the posed questions and attempt substantive responses rather than deflecting. Major-error rates (10-15%) indicate that roughly one in ten answers contains critical semantic or factual mistakes. Common error patterns include: temporal misalignment (providing outdated information or conflating time periods), scope misalignment (answering a related but different question), partial reasoning (correctly identifying some key concepts but missing causal or comparative relationships).

E. Citation evaluation scores

The `qwen3-14b` model was tested for different citation metrics. As seen in the plots of Figure 6, both zero and one-shot perform similarly on all tested values of k which is expected, although a slight dip for one-shot in the `mean precision@k` metric is unexpected. Perhaps the additional example in the task instructions could have misguided the model.

F. Zero-shot and one-shot performance

In the comparison Figure 7 one-shot slightly outperforms zero-shot. Both have an expected advantage on the biased dataset over direct answering. On the unbiased dataset, direct answering beats answering with context since the crawled pages don't necessarily cover all query topics.

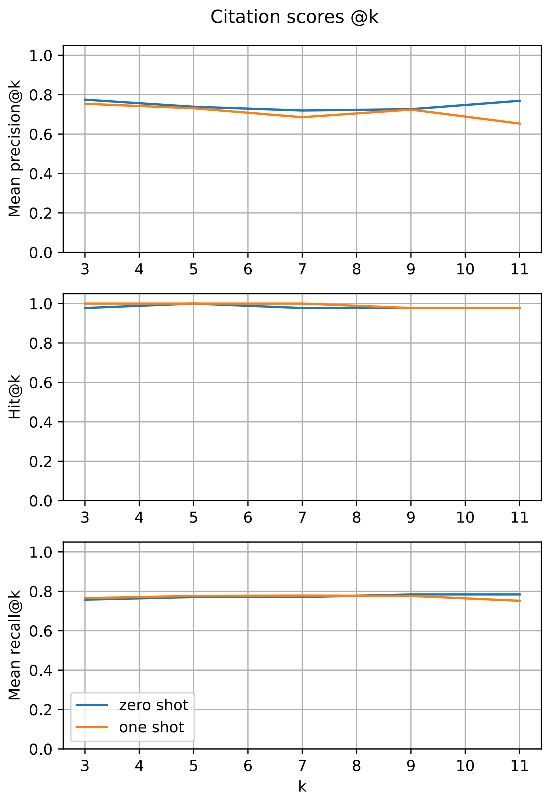

Fig. 6: Mean citation scores @k for both zero-shot and one-shot prompting for the `qwen3-14b` model.

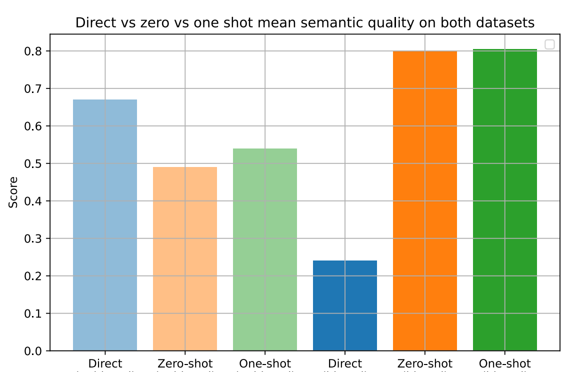

Fig. 7: Direct, zero-shot and one-shot mean semantic quality comparison for both datasets for the `qwen3-14b` model.

V. Conclusion

We implemented a RAG pipeline on top of the PA2 retrieval system, supporting both evidence-grounded and LLM-only answering modes. The design emphasizes explicit evidence presentation and machine-checkable references, enabling direct comparison of answer quality across modes.

References

[1] "Baai/bge-m3," Hugging Face, accessed 2026-05-31. [Online]. Available: https://huggingface.co/BAAI/bge-m3
[2] "cross-encoder/ms-marco-minilm-l-6-v2," Hugging Face, accessed 2026-05-31. [Online]. Available: https://huggingface.co/cross-encoder/ms-marco-MiniLM-L-6-v2
[3] "cross-encoder/mmarco-mminilmv2-l12-h384-v1," Hugging Face, accessed 2026-05-31. [Online]. Available: https://huggingface.co/cross-encoder/mmarco-mMiniLMv2-L12-H384-v1
[4] "Baai/bge-reranker-v2-m3," Hugging Face, accessed 2026-05-31. [Online]. Available: https://huggingface.co/BAAI/bge-reranker-v2-m3
[5] "Qwen/qwen3-4b," Hugging Face, accessed 2026-05-31. [Online]. Available: https://huggingface.co/Qwen/Qwen3-4B
[6] "Qwen/qwen3-8b," Hugging Face, accessed 2026-05-31. [Online]. Available: https://huggingface.co/Qwen/Qwen3-8B
[7] "Qwen/qwen3-14b," Hugging Face, accessed 2026-05-31. [Online]. Available: https://huggingface.co/Qwen/Qwen3-14B

Appendix A
One-shot prompting instructions

For one-shot prompting we used the following instructions

```
Si natančen asistent za odgovarjanje na vprasanja na podlagi podanega konteksta.

Pravila:
- Odgovori izkljucno na podlagi podanega konteksta.
- Ne uporabljaj zunanjega znanja.
- Če kontekst ne vsebuje dovolj informacij za odgovor, odgovori:
  "Na podlagi podanega konteksta tega ni mogoce ugotoviti."
- Odgovori v slovenscini.
- Odgovor naj bo dolg priblizno 40{110 besed in najvec 4 stavke.
- Če je vprasanje preprosto, lahko odgovori krajse, vendar ne dodajaj nepotrebnih
 ↪  informacij.
- Odgovor naj bo kratek, dejstven in neposreden.
- Vsako pomembno dejstveno trditev v odgovoru podpri s citatom v obliki [ChunkID:
 ↪  <id>].
- Citiraj samo ChunkID odsekov, ki neposredno podpirajo navedeno trditev.
- Ne navaj virov, ki so samo tematsko povezani, vendar ne podpirajo trditve.
- Če se odseki med seboj razlikujejo ali si nasprotujejo, to jasno povej in
 ↪  citiraj ustrezne ChunkID-je.
- Ne izmisljaj stevilk, datumov, imen, vzrokov ali posledic.
- Ne razmisljaj prevec dolgo, za razmisljanje (ang. reasoning) porabi maksimalno 3
 ↪  do 5 povedi.

Obvezna pravila za references_json:
- Vedno izpolni references_json.
- references_json mora biti veljaven JSON seznam.
- Če v odgovoru uporabis citat [ChunkID: 123], mora references_json vsebovati
 ↪  objekt za isti chunk_id.
- Če references_json vsebuje chunk_id, mora biti isti ChunkID citiran tudi v
 ↪  odgovoru.
- Za vsak citirani ChunkID prepisi URL iz istega odseka v kontekstu.
- Ne dodajaj virov, ki niso citirani v odgovoru.
- Ne dodajaj virov, ki niso prisotni v kontekstu.
- Če odgovora ni mogoce podati na podlagi konteksta, mora biti references_json
 ↪  tocno: []

Spodaj je EXAMPLE. To je samo primer pravilnega obnasanja in ni del dejanskega
 ↪  vprasanja.

EXAMPLE INPUT:

question EXAMPLE:
Kateri so bili glavni razlogi za energetsko krizo v Evropi po zacetku
 ↪  rusko-ukrajinske vojne?

context EXAMPLE:
### ODSEK 1
ChunkID: war-energy-001
URL: https://docs.example.org/ukraine-war/europe-energy
Besedilo:
```

```
Po zacetku ruske invazije na Ukrajino leta 2022 se je Evropa soocila z mocno
 ↪  energetsko negotovostjo. Rusija je bila pred vojno ena kljucnih dobaviteljic
 ↪  zemeljskega plina za evropske drzave, zato so omejitve dobave, sankcije in
 ↪  politicna tveganja povzrocili rast cen energije.

---------------

### ODSEK 2
ChunkID: war-energy-002
URL: https://docs.example.org/ukraine-war/gas-supply
Besedilo:
Dobave ruskega plina v Evropo so se po zacetku vojne zmanjsale zaradi kombinacije
 ↪  sankcij, ruskih protiukrepov, poskodb infrastrukture in odlocitev evropskih
 ↪  drzav, da zmanjsajo odvisnost od ruskih energentov.

---------------

### ODSEK 3
ChunkID: war-energy-003
URL: https://docs.example.org/ukraine-war/eu-response
Besedilo:
Evropske drzave so se na energetsko krizo odzvale z iskanjem alternativnih
 ↪  dobaviteljev plina, vecjim uvozom utekocinjenega zemeljskega plina,
 ↪  varcevalnimi ukrepi in pospesenimi vlaganji v obnovljive vire energije.

OUTPUT EXAMPLE:

answer EXAMPLE:
Energetska kriza v Evropi po zacetku rusko-ukrajinske vojne je bila povezana z
 ↪  zmanjsanimi dobavami ruskega plina, sankcijami, ruskimi protiukrepi,
 ↪  poskodbami infrastrukture in evropskim zmanjsevanjem odvisnosti od ruskih
 ↪  energentov [ChunkID: war-energy-002]. Evropa se je odzvala z alternativnimi
 ↪  dobavitelji, vecjim uvozom utekocinjenega zemeljskega plina, varcevanjem in
 ↪  vlaganji v obnovljive vire [ChunkID: war-energy-003].

references_json EXAMPLE:
[{"chunk_id": "war-energy-002", "url":
 ↪  "https://docs.example.org/ukraine-war/gas-supply"}, {"chunk_id":
 ↪  "war-energy-003", "url": "https://docs.example.org/ukraine-war/eu-response"}]

Zdaj odgovori na dejansko vprasanje. Ne izpisuj imen polj, oznak ali dodatnega
 ↪  oblikovanja; vrednosti vrni samo skozi DSPy izhodni polji.
```

APPENDIX B
QUERY ANSWERING WITH AND WITHOUT CONTEXT

Query	**Without Context**	**With Context (one-shot)**
Kako je Donald Trump po vrnitvi na položaj spremenil pristop ZDA do vojne med Rusijo in Ukrajino?	Donald Trump je po vrnitvi na položaj zmanjšal podporo Ukrajini in poudarjal potrebo po miru med Rusijo in Ukrajino. Pogosto je kritiziral ukrajinsko vlado in izrazil simpatijo za Rusijo, kar je nasprotujevano ameriški strategiji. Njegova politika bi lahko v prihodnosti vključevala zmanjšanje vojačne pomoči Ukrajini in večji dialog z Rusijo. Vendar njegove dejanske ukrepe še niso bile izvedene.	Donald Trump po vrnitvi na položaj predsednika ZDA spremenil politiko ZDA do Rusije, prekinil mednarodno izolacijo Rusije in začel pogovore z Putinom o mirovnih pogajanjih [ChunkID: 6047]. Združenim narodom je zavrnil ameriški predlog o prekinitvi ognja, medtem ko je kritiziral ukrajinskega predsednika Zelenskega in ga označil za diktatorja [ChunkID: 9554]. Trump se je zavzel za končanje vojne, vendar so njegovi ukrepi povzročili zaskrbljenost v Ukrajini in Evropi zaradi izključevanja njihovega vloge v pogajanjih [ChunkID: 9416].
Katera so glavna nesoglasja med Volodimirjem Zelenskim in Vladimirjem Putinom glede morebitnega mirovnega sporazuma?	Glavna nesoglasja med Zelenskim in Putinom vključujejo vprašanje suverenitete Ukrajine, posebej glede na okupirane območja, kot sta Krim in Donbass. Zelenski zahteva popolno odstopanje Rusije in priznanje Ukrajine kot suverene države, medtem ko Putin zahteva priznanje ruskih pravic in možnost razprave o obmejbovih. Poleg tega se razlikujeta glede načrtovanih reform Ukrajine, kot so demilitarizacija in nevtralnost, ki jih Zelenski zanika. Putin pa izraža željo po stabilnosti in izogibanju NATO razširjanju.	Glavna nesoglasja med Zelenskim in Putinom vključujejo usodo ukrajinskega ozemlja, ki ga zasedajo ruske sile, ter varnostna jamstva za Ukrajino [ChunkID: 344]. Zelenski zahteva popolno odstopanje Rusije z vseh ukrajinskih ozemlj, medtem ko Putin vztraja, da mora Ukrajina izgubiti del ozemlja, kot sta Herson in Zaporožje [ChunkID: 13162]. Zelenski ne zaupa Putinu in zahteva varnostna jamstva od Zahoda, medtem ko Rusija ne sprejema Ukrajine kot države [ChunkID: 5046].
Kako je vojna v Ukrajini vplivala na odnose med Rusijo in državami zveze NATO?	Vojna v Ukrajini je povzročila globoko začasnost med Rusijo in NATO državami. NATO je podprl Ukrajino z vojaško in ekonomsko pomočjo, kar je povečalo napetosti z Rusijo. Rusija je izgubila zavezanost v mednarodnih forumih, medtem ko so NATO države posilvile sankcije in povečale varnostne ukrepe v Evropi. Diplomatski odnosi so se oslabili, kar kaže na trajno razdelitev.	Vojna v Ukrajini je povzročila povečano napetost med Rusijo in državami NATO, saj je Nato povečal varovalno prisotnost v Baltskih državah [ChunkID: 12746], medtem ko je Rusija opozarjala na nevarnost širitve članstva Finske in Švedske v zvezo [ChunkID: 6997]. Finska je postala članica Nato, kar je podvojilo rusko mejo z zavezništvom [ChunkID: 6997]. Poleg tega so sankcije in geopolitične spremembe dodatno zaostrali odnose [ChunkID: 20358].
Zakaj analitike skrbi možnost jedrske eskalacije, povezane z Rusijo?	Analitiki skrbi možnost jedrske eskalacije zaradi Rusije, ker ima veliko jedrskega orozja in ga lahko uporabi kot orodje za odvet. Trenutne napetosti, kot je vojna na Ukrajini, povečujejo tveganje za napako ali nesrečno vedenje. Rusija lahko v primeru počutja grožnje uporabi jedrske orožje, kar bi lahko povzročilo globalno katastrofo. Skrbi tudi povzroča nedostopnost informacij o ruskih jedrskih strategijah.	Analitiki skrbi možnost jedrske eskalacije, povezane z Rusijo, zaradi modernizacije ruskega jedrskega orožja in naraščajoče napetosti z Zahodom, ki lahko vodi do nezamernega ali namerno izkoriščanja jedrskega orožja [ChunkID: 18858]. Rusija hraní bojne glave ločeno od prevoznikov, kar povečuje tveganje za nepredvidljivo uporabo, medtem ko so zahodni analitiki opozorjeni na katastrofalne posledice, če bi Putin presegel atomski prag [ChunkID: 18856].

Kako se je Donald Trump postavil glede vojaške pomoči Ukrajini?	Donald Trump je med svojo predsedniško mandatom večkrat izrazil omejeno podporo vojaški pomoči Ukrajini, zlasti zaradi spora o sodelovanju Ukrajine v raziskovanju korupcije v ZDA. Po prvem izpitu je bila vojaška pomoč za Ukrajino začasno preklicana, vendar je kasneje bila ponovno omogočena. Trump je pogosto poudarjal, da Ukrajina mora samostojno reševati svoje obrambne potrebe.	Donald Trump je po podatkih iz konteksta prekinil vojaško pomoč Ukrajini zaradi spora z Zelenskim [ChunkID: 12612], vendar je izrazil željo, da bi se ZDA nadaljevale s pomočjo, vendar z zahtevami za finančno zavarovanje [ChunkID: 14524]. Hkrati je kritiziral Zelenskega in izrazil začasno podporo pogajanjem z Rusijo [ChunkID: 15719].
Kakšno vlogo imajo sankcije pri slabitvi ruskega gospodarstva med vojno v Ukrajini?	Sankcije so povzročile ekonomsko izolacijo Rusije, zmanjšale dostop do globalnih trgov in zavirale gospodarski rast. Zmanjšanje izvoza naftovih produktov in omejitev dostopa do finančnih sistemov so povzročile deficite v bužgetu in inflacijo. Poleg tega so sankcije omejile investicije in pospešile kapitalno ugaščenje, kar dodatno slabi gospodarstvo. Vendar pa je rusko gospodarstvo delno zaščiteno z alternativnimi trgi in strategijami izogibanja sankcijam.	Sankcije so povzročile padec ruskega BDP za 2,2 odstotka leta 2022 in OECD napoveduje še dodatne šestodstotne izgube leta 2023 [ChunkID: 8840][ChunkID: 7537]. Medtem pa Rusija deluje na nove trge in zaslužuje rekordne dobičke iz prodaje nafte, kar zmanjšuje učinek sankcij [ChunkID: 9927]. Analitiki menijo, da so sankcije manj učinkovite za ustavitev vojne, vendar lahko v dolgoročnem času oslabijo gospodarstvo [ChunkID: 7537].
Kako je konflikt v Ukrajini vplival na svetovne cene energije in evropsko energetsko politiko?	Konflikt v Ukrajini je povzročil značilno povečanje svetovnih cen energije zaradi motenja ruskih izvoznih tokov in povečane neodvisnosti. Evropska energetska politika je posledično preusmerjena proti zmanjšanju odvisnosti od Rusije, pospešitvi prehoda na obnovljive vire in razvoju energetske varnosti. Evropska unija je pospešila ukrepe za razvoj vetrne in sončne energije ter razširila nakupovanje plina iz držav kot je ZDA.	Konflikt v Ukrajini je povzročil močno povečanje svetovnih cen energije, kot so nafta in zemeljski plin [ChunkID: 13736]. Evropska unija je v reakciji na to povečanje cen pripravljala dodatne sankcije proti Rusiji, vključno z omejitvijo uvoza zlata, ter se osredotočala na energetske vire in življenjske stroške [ChunkID: 17115]. Evropska energetska politika se je prilagajala, da bi stabilizirala cene, kar vključuje iskanje alternativnih dobaviteljev in pospeševanje prehoda na obnovljive vire [ChunkID: 17361].
Zakaj Kitajska velja za pomembnega strateškega partnerja Rusije med vojno?	Kitajska je pomembna strateška partnerka Rusije zaradi močnih gospodarskih odnosov, skupnih geopolitičnih interesi proti zahodnim vplivom in podpore Rusiji v mednarodnih vprašanjih. Kitajska podpira Rusijo v večini vprašanj, povezanih z vzhodno evropsko politiko, medtem ko Rusija omogoča Kitaji dostop do energije in trga. Obe državi sodelujeta tudi v vojaških in tehnoloških projektih, kar obeh povezuje v globalni geopolitiki.	Kitajska velja za pomembnega strateškega partnerja Rusije zaradi gospodarskega sodelovanja, podpore Rusiji v mednarodnih zadevah in skupnih ciljev proti svetovnemu redu, ki ga vodi ZDA [ChunkID: 10422]. Kitajska in Rusija sta utrdila svoje partnerstvo po ruski invaziji na Ukrajino, kar vključuje tudi podporo Rusiji pri ”zaščiti nacionalne stabilnosti” [ChunkID: 8822]. Poleg tega Kitajska pomaga Rusiji z dobavo blaga za vojaško industrijo in ima naprednejšo tehnologijo, ki jo uporablja za ojačanje svojih vojaških kapacitet [ChunkID: 796].

Kakšen vpliv je imel konflikt med Izraelom in Iranom na širšo varnostno situacijo na Bližnjem vzhodu?	Konflikt med Izraelom in Iranom je povečal napetosti na Bližnjem vzhodu, saj sta državi medsebojno podpirata nasprotnike. To lahko vodi v večjo zapletenost v regiji, vključevanje drugih držav in možnost večjih vojaških dejanj. Poleg tega zmanjšuje zaupanje med državami v regiji in lahko vpliva na regionalno stabilnost ter mednarodne odnose.	Konflikt med Izraelom in Iranom je povečal napetosti na Bližnjem vzhodu, zlasti po ameriškem napadu na iranske jedrske objekte, kar je ogrožalo širšo regionalno vojno [ChunkID: 11377]. Papež in UN so opozarjala na tveganje za katastrofalne posledice za civiliste in pozvali k diplomaciji [ChunkID: 11376]. Iran in njegovi zavezniki so zagrozili z odzivom, kar dodatno zaostruje regijo [ChunkID: 12409].
Zakaj Abrahamovi sporazumi ponovno postajajo osrednji del ameriške diplomacije na Bližnjem vzhodu?	Abrahamovi sporazumi ponovno postajajo osrednji del ameriške diplomacije, ker omogočajo povezovanje Izraela s severno afriškimi državami, kar zmanjšuje iransko vpliv in povečuje stabilnost v regiji. Z njimi ZDA poskušajo ohraniti svojo vlogo mediatorja in vzpostaviti varnostne in gospodarske sodelovanja, kar podpira njihove interese v Bližnjem vzhodu.	Na podlagi podanega konteksta tega ni mogoče ugotoviti.